Flip Flop



SAMARTH

CUET PG 2027

Lecture – 14

Unit 3 : Digital Fundamental
Topic : Flip Flops

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Recap of previous lecture

1. ~~Registers~~ Sequential circuits

SR latches.

SR flip flop.



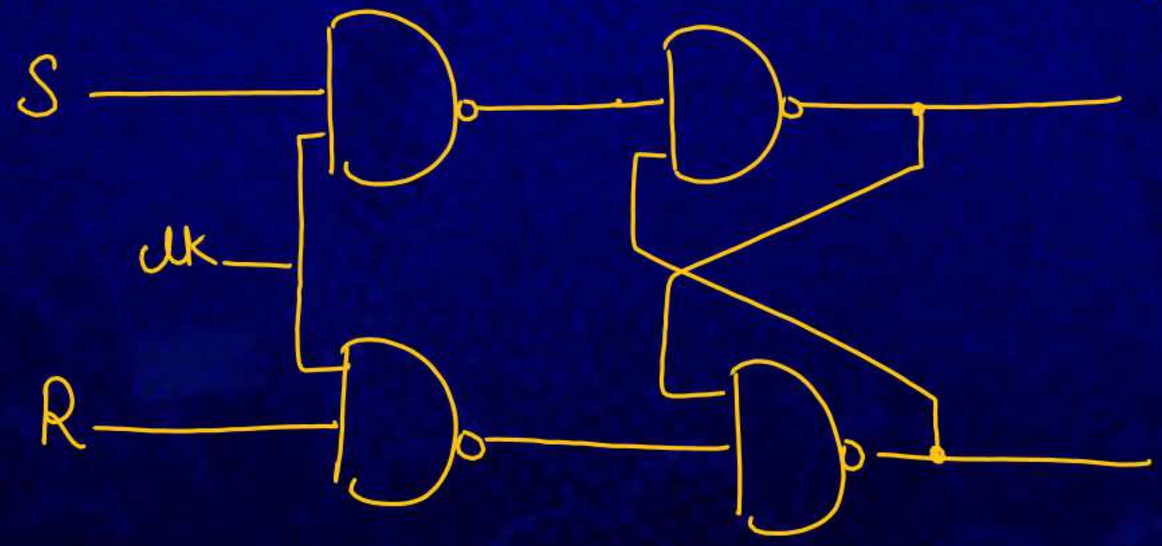
Topics *to be covered*

1. Flip Flops



Topic : Flip Flops

SR Flip Flop



S R flip flop

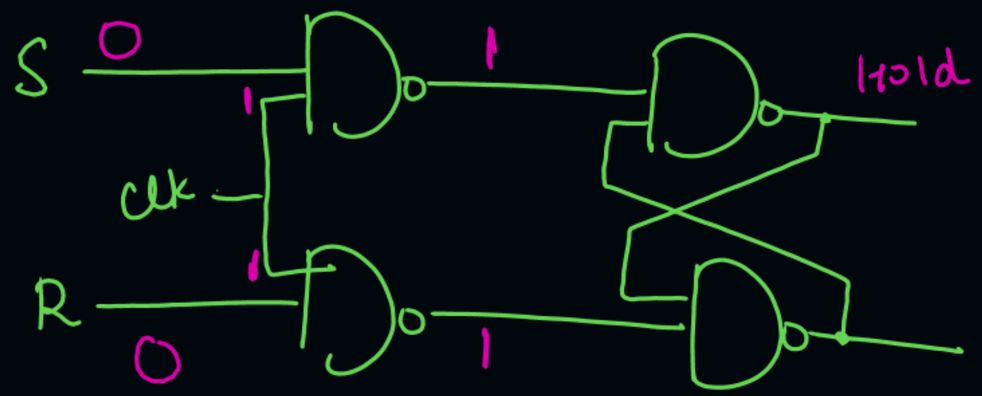
Clock	S	R	Q_{n+1}
Not Triggered 0	X	X	Q_n
Triggered ↑	0	0	Hold state
	0	1	0
	1	0	1
	1	1	Invalid

SR latch

S	R	
0	0	— Invalid
0	1	— 1 ✓
1	0	— 0 ✓
1	1	— Hold state

① $S=0, R=0$

Clock = 1

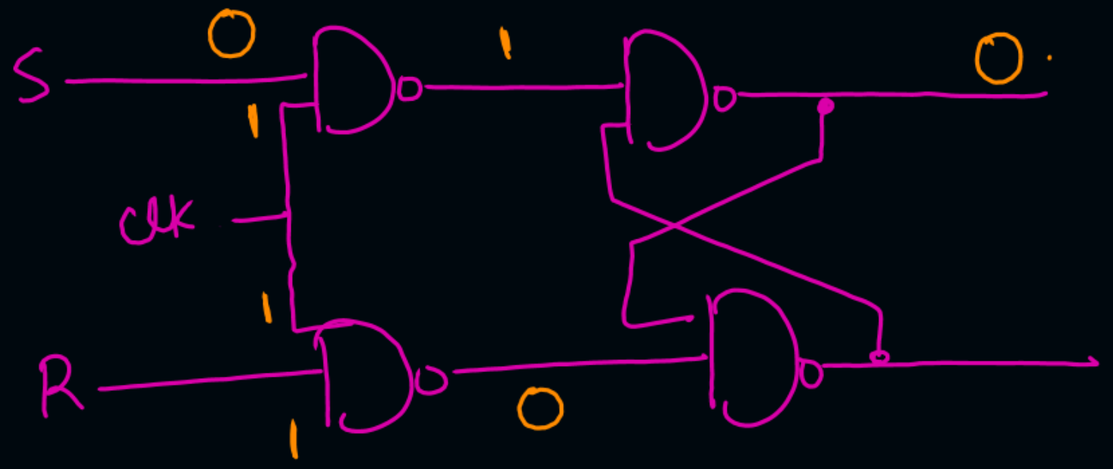


NAND

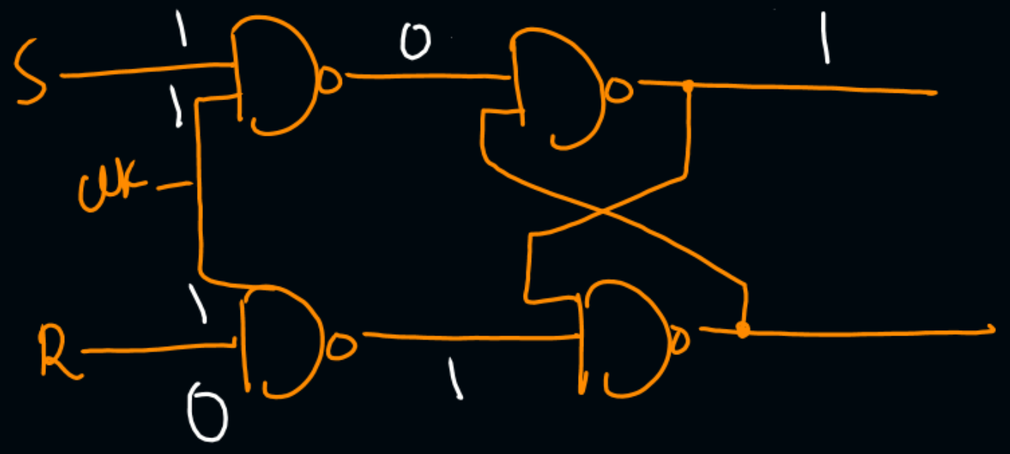
A	B	A·B	$\overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

If any one input is 0, then o/p is 1

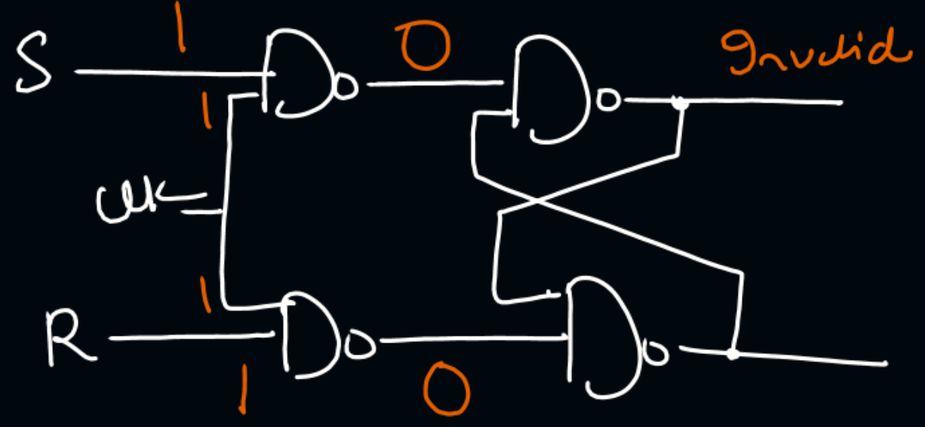
② $S=0, R=1$ Clock = 1.



③ $S=1, R=0,$ Clock = 1



(4) $S = 1, R = 1$.



Characteristic eq

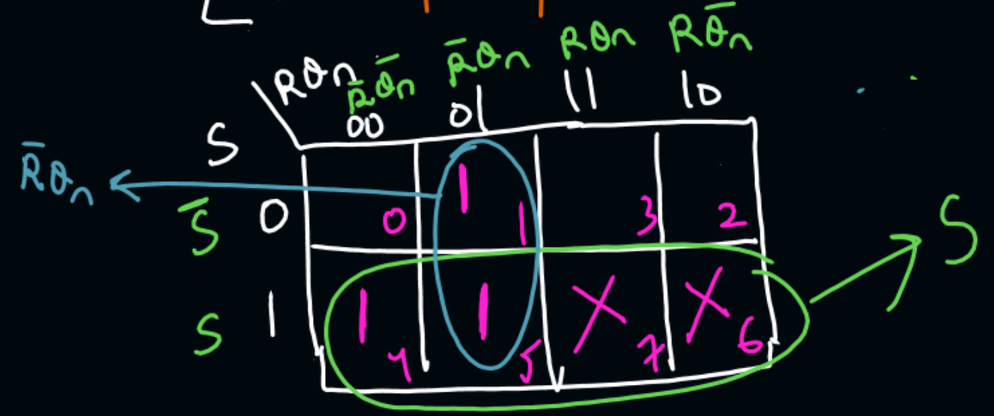
$$\bar{R}Q_n + S$$

Ans.

Characteristic Table

	S	R	Q_n	Q_{n+1}
Hold	0	0	0	0
	0	0	1	1
Reset	0	1	0	0
	0	1	1	0
Set	1	0	0	1
	1	0	1	1
Invalid	1	1	0	X
	1	1	1	X

Don't care



Toggle (complement)

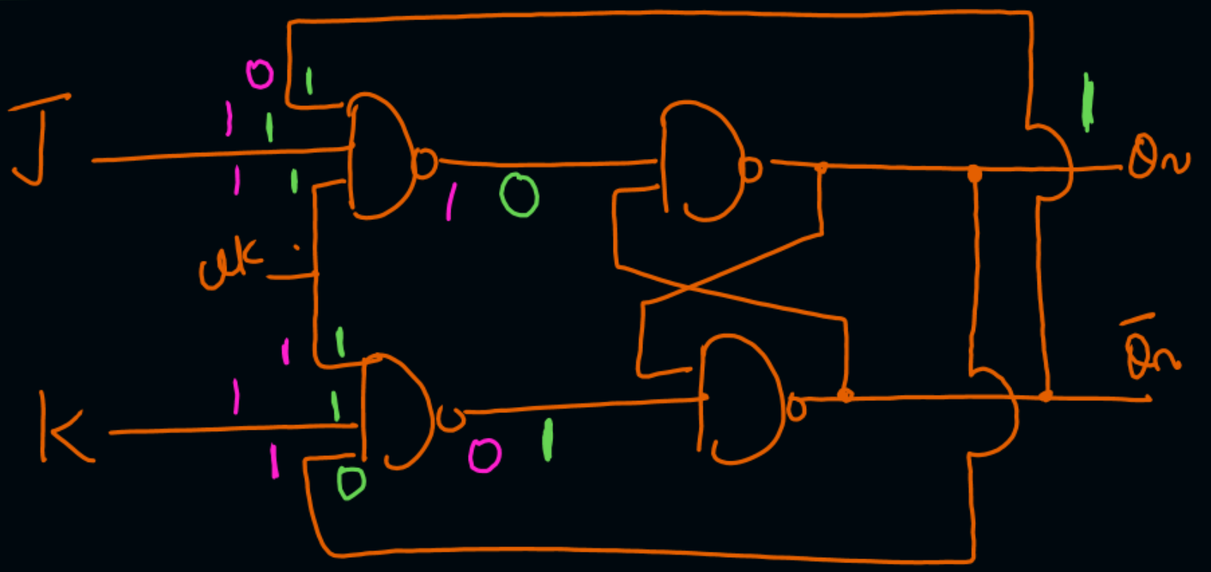
Jk flip flop.

It is same as SR flip flop

SR	O/P
0 0	Hold
0 1	0
1 0	1
1 1	Invalid.

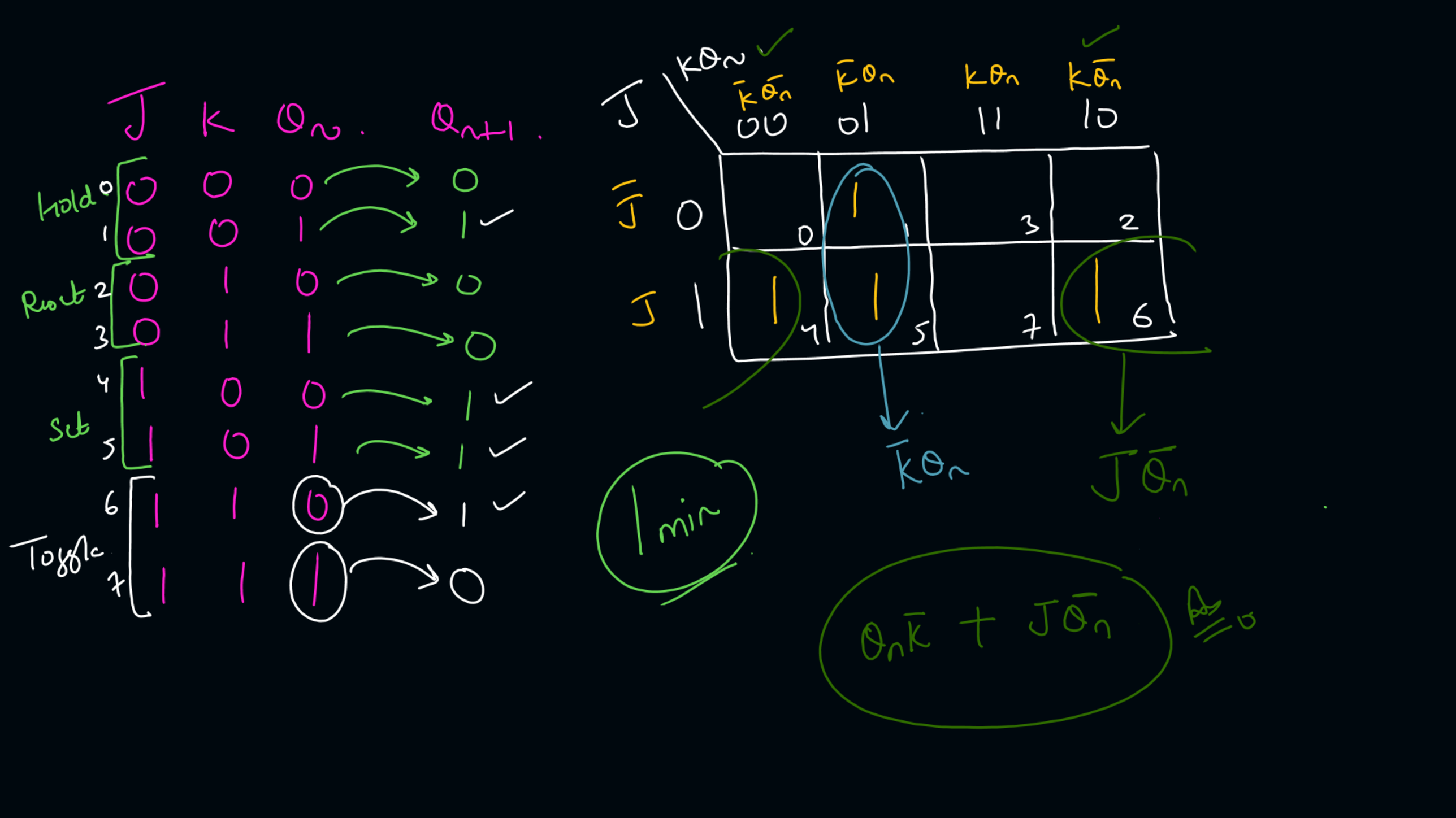
Same in SR & JK

↳ We will make small change in design to solve this problem.



	J	K	Q_n	$\bar{Q}_n$	
①	1	1	0	1	— Also 1 Toggle Q_n
②	1	1	1	0	— Also 0 Toggle Q_n

J	K	Q_{n+1}
0	0	Hold
0	1	0
1	0	1
1	1	$\bar{Q}_n$

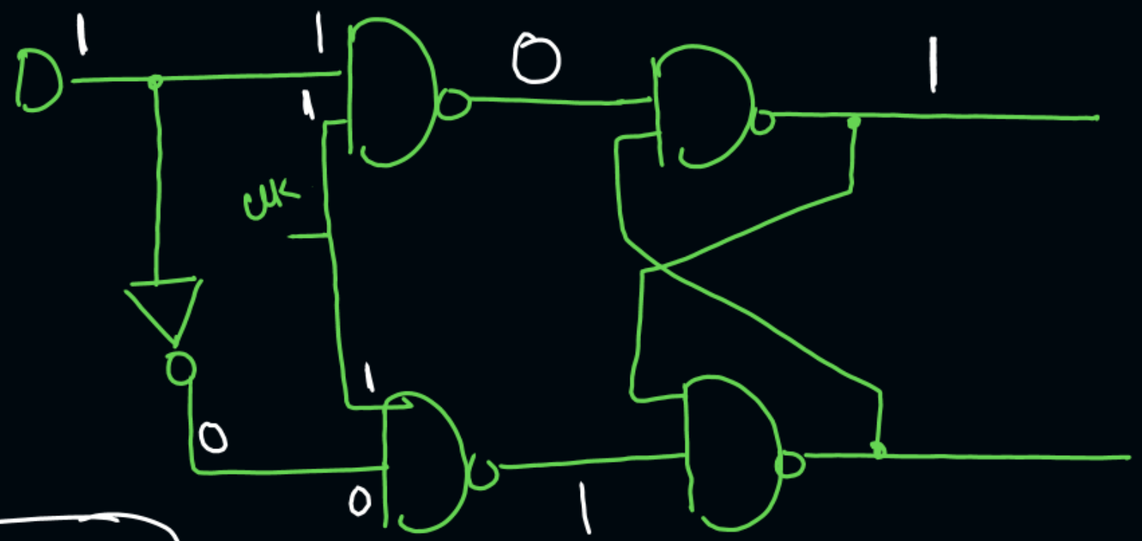

J K Qn Qn+1
Hold
Reset
Set
Toggle
0
1
2
3
4
5
6
7
KQn
K̄Q̄n
K̄Qn
KQn
KQ̄n
J
J̄
0
1
0
1
3
2
4
5
7
6
K̄Qn
JQ̄n
1 min
QnK̄ + JQ̄n

D flip flop.

(Called as storage device)

(Here o/p is same as q/p)

D	Q_{n+1}
0	0
1	1



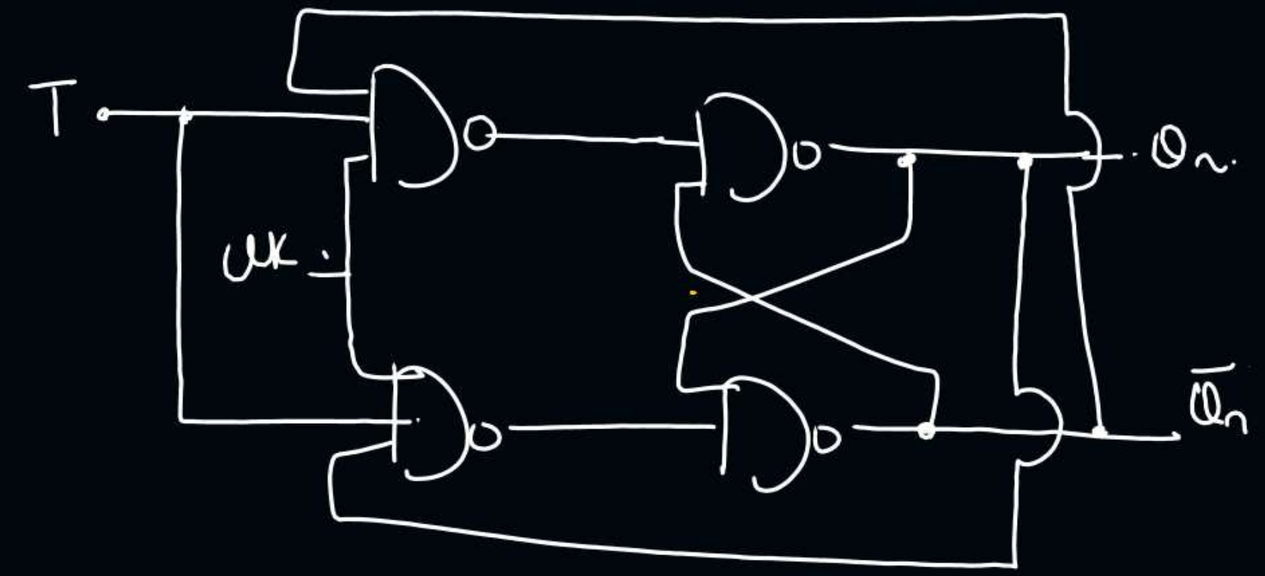
$D = Q_{n+1}$

T (Toggle) Flip Flop.

Single input version of JK flip flop.

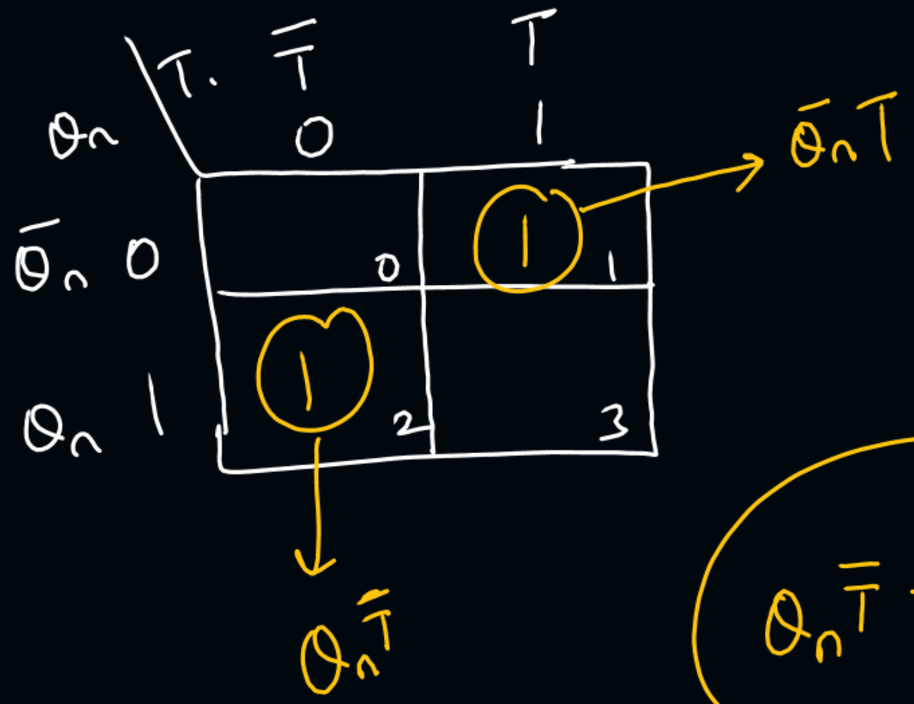
Q(n)	T	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0

When T = 1
Then Q_{n+1} is complement Q_n.



T	
0	Q_n
1	$\overline{Q_n}$

	Q_n	T	Q_{n+1}
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0



$$Q_n\bar{T} + \bar{Q}_n T$$

$$\boxed{Q_n \oplus T}$$

PYQ 2025.

	J	K	Next state
A	0	0	1 Toggle
B	0	1	2 Set
C	1	0	3 Reset
D	1	1	4 No change

(1) A-1, B-2, C-3, D-4

(2) A-4, B-3, C-2, D-1

(3) A-4, B-3, C-1, D-2

(4) A-1, B-2, C-4, D-3

Topic : Flip Flops

1. Introduction to Flip-Flops

A Flip-Flop is a basic sequential logic circuit used to store one bit of binary information, either 0 or 1.

Unlike combinational circuits, the output of a flip-flop depends not only on the present input but also on its previous state.



Topic : Flip Flops

Key Features:

- Stores 1 bit of information.
- Has two stable states: 0 and 1.
- Used as a basic memory element.
- Usually controlled by a clock signal.
- Forms the basic building block of registers, counters, memory units, and sequential circuits.



2. Flip-Flop vs Latch

Basis	Latch	Flip-Flop
Triggering	Level-triggered	Usually edge-triggered
Control	Enable signal	Clock signal
Operation	Active during a clock level	Active at a clock edge
Speed	Generally faster	Comparatively controlled
Application	Temporary storage	Registers, counters, memory

Important:

A flip-flop generally changes its state only at a specified clock edge, whereas a latch can respond throughout the active level of the control signal.



3. Types of Flip-Flops

The four important types are:

1. SR Flip-Flop
2. JK Flip-Flop
3. D Flip-Flop
4. T Flip-Flop



4. SR Flip-Flop

The SR (Set-Reset) Flip-Flop has two inputs:

- S = Set
- R = Reset

It has two outputs:

- Q
- $\overline{Q}$



SR Flip-Flop Truth Table

S	R	Next State Q(t+1)	Operation
0	0	Q(t)	No Change
0	1	0	Reset
1	0	1	Set
1	1	Invalid	Forbidden condition



Important Points

- S = 1, R = 0 → Set
- S = 0, R = 1 → Reset
- S = 0, R = 0 → No change
- S = 1, R = 1 → Invalid/forbidden state for the conventional active-high SR implementation.



Characteristic Equation

$$Q(t+1) = S + \bar{R}\,Q(t)$$

The equation is valid subject to the restriction that S and R cannot both be 1.



5. JK Flip-Flop

The JK Flip-Flop is an improved version of the SR flip-flop.

It has two inputs:

- J
- K

The major advantage is that it eliminates the invalid state of the SR flip-flop.



Topic : Flip Flops

JK Flip-Flop Truth Table

J	K	Next State Q(t+1)	Operation
0	0	Q(t)	No Change
0	1	0	Reset
1	0	1	Set
1	1	$\overline{Q(t)}$	Toggle



Important Points

When:

$$J = K = 1$$

the flip-flop toggles its state.

For example:

$$0 \rightarrow 1$$

and

$$1 \rightarrow 0$$

Characteristic Equation

$$Q(t+1) = J\,\bar{Q}(t) + \bar{K}\,Q(t)$$



6. Race-Around Condition

The race-around condition is associated particularly with a level-triggered JK flip-flop when:

$$J = K = 1$$

and the clock pulse remains active long enough for the output to toggle repeatedly.

Example:
If the initial state is:

$$Q = 0$$

then during the active clock pulse:

$$0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow \cdots$$

Thus, the final state becomes uncertain.



Methods to Avoid Race-Around Condition

1. Edge-triggered flip-flop
2. Master-Slave JK flip-flop
3. Using a sufficiently narrow clock pulse

Point:

The Master-Slave JK flip-flop is commonly used to eliminate the race-around problem.



7. D Flip-Flop

The D (Data/Delay) Flip-Flop has only one input:

- D = Data

Its next state directly follows the input D at the triggering clock edge.

Truth Table

D	Q(t+1)	Operation
0	0	Reset
1	1	Set

Therefore:

$$Q(t+1) = D$$



Topic : Flip Flops

Main Advantage:

The D flip-flop has no invalid input combination.

Applications:

- Data storage
- Registers
- Shift registers
- Memory circuits
- Synchronization circuits

Important Fact:

$$Q(t+1) = D$$



8. T Flip-Flop

The T (Toggle) Flip-Flop has a single input:

- T = Toggle

Truth Table:

T	Q(t+1)	Operation
0	Q(t)	No Change
1	$\overline{Q(t)}$	Toggle



Characteristic Equation:

$$Q(t+1) = T \oplus Q\,(t)$$

where $\oplus$ represents the XOR operation.



Important Application:

T flip-flops are widely used in binary counters and frequency division circuits.

If T = 1 continuously, the output toggles at every active clock edge.

Thus, ideally:

$$f_{out} = \frac{f_{clock}}{2}$$



9. Comparison of Flip-Flops

Feature	SR	JK	D	T
Inputs	S, R	J, K	D	T
No Change	00	00	—	0
Set	10	10	D=1	—
Reset	01	01	D=0	—
Toggle	—	11	—	1
Invalid State	Yes	No	No	No
Main Application	Basic storage	Counters/sequential circuits	Data storage	Counters



10. Excitation Tables

An excitation table specifies the required inputs to change the flip-flop from the present state to the desired next state.

SR Flip-Flop Excitation Table

Q(t)	Q(t+1)	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Here, X = Don't Care.



Topic : Flip Flops

JK Flip-Flop Excitation Table

Q(t)	Q(t+1)	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



D Flip-Flop Excitation Table

Q(t)	Q(t+1)	D
0	0	0
0	1	1
1	0	0
1	1	1

Therefore:

$$D = Q(t+1)$$



T Flip-Flop Excitation Table

Q(t)	Q(t+1)	T
0	0	0
0	1	1
1	0	1
1	1	0



11. Characteristic Equations

The characteristic equations are extremely important for competitive examinations.

Flip-Flop	Characteristic Equation
SR	$Q(t+1) = S + \bar{R}\,Q(t)$
JK	$Q(t+1) = J\,\bar{Q(t)} + \bar{K}\,Q(t)$
D	$Q(t+1) = D$
T	$Q(t+1) = T \oplus Q\,(t)$



12. Conversion of Flip-Flops

One type of flip-flop can be converted into another by using appropriate combinational logic.

JK $\rightarrow$ T

For a JK flip-flop:

$$J = T$$

$$K = T$$



Thus, connecting J and K together gives T-flip-flop behavior.

D → T

$$D = T \oplus Q\,(t)$$

JK → D

$$J = D$$

$$K = \bar{D}$$

SR → D

A common conversion is:

$$S = D, R = \bar{D}$$

This avoids the invalid $S = R = 1$condition.



13. Edge Triggering

A flip-flop may be designed to respond to either:

- Positive/rising edge: $0 \rightarrow 1$
- Negative/falling edge: $1 \rightarrow 0$

Positive Edge:

The flip-flop changes state at the transition:

$$0 \rightarrow 1$$



Negative Edge:

The flip-flop changes state at:

$$1 \rightarrow 0$$

Point:

Edge-triggered flip-flops are useful because they restrict state changes to a particular clock transition.



14. Master-Slave Flip-Flop

A Master-Slave Flip-Flop consists of two stages:

1. Master
2. Slave

The two stages operate on opposite clock levels.

Purpose:

The main purpose is to prevent race-around conditions, especially in JK flip-flops.



15. Applications of Flip-Flops

Flip-flops are fundamental components of sequential digital systems.

Major applications:

- Registers
- Shift registers
- Counters
- Frequency dividers
- Memory units
- Data storage
- Sequence generators
- Control circuits
- Digital clocks
- State machines



16. Important Facts

1. A flip-flop stores one bit of information.
2. The D flip-flop is commonly called a data or delay flip-flop.
3. The JK flip-flop removes the forbidden state of the conventional SR flip-flop.
4. When J = K = 1, a JK flip-flop toggles.
5. When T = 1, a T flip-flop toggles.
6. T flip-flops are commonly used in counters.
7. The characteristic equation of D flip-flop is:

$$Q(t+1) = D$$



Topic : Flip Flops

8. The characteristic equation of T flip-flop is:

$$Q(t+1) = T \oplus Q\,(t)$$

9. The race-around condition is associated with a level-triggered JK flip-flop when $J = K = 1$.
10. A Master-Slave JK flip-flop can eliminate the race-around condition.
11. A flip-flop is a sequential circuit, whereas a basic combinational circuit has no memory of previous states.
12. A positive-edge-triggered flip-flop responds to the rising edge of the clock.



Quick Memory Trick

SR → Set/Reset

JK → No invalid state + Toggle

D → Data follows Q

T → Toggle



2 Mins Summary

↳ Flip flops-

Homework from SAMARTH CUET PG 2027 Module

↳ Revision
+
DPP

THANK YOU